

Week 3

Session 11:

Date and Time Handling Concepts-

1. Java Date and Time –

Java code:

```
import java.io.*;
import java.util.*;

public class Solution {

    public static String findDay(int month, int day, int year) {

        Calendar cal = Calendar.getInstance();

        cal.set(year, month - 1, day);

        String dayName = cal.getDisplayName(

            Calendar.DAY_OF_WEEK,

            Calendar.LONG,

            Locale.ENGLISH

        );

        return dayName.toUpperCase();

    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int month = sc.nextInt();

        int day = sc.nextInt();

        int year = sc.nextInt();

        System.out.println(findDay(month, day, year));

    }

}
```

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Sample Input

08 05 2015

Sample Output

WEDNESDAY

2. Number of Days Between Two Dates –

Java code:

```
class Solution {  
    public int daysBetweenDates(String date1, String date2) {  
        String[] d1 = date1.split("-");  
        String[] d2 = date2.split("-");  
        int y1 = Integer.parseInt(d1[0]);  
        int m1 = Integer.parseInt(d1[1]);  
        int day1 = Integer.parseInt(d1[2]);  
        int y2 = Integer.parseInt(d2[0]);  
        int m2 = Integer.parseInt(d2[1]);  
        int day2 = Integer.parseInt(d2[2]);  
        return Math.abs(toDays(y1, m1, day1) - toDays(y2, m2, day2));  
    }  
    private int toDays(int year, int month, int day) {  
        int[] days = {  
            31, 28, 31, 30, 31, 30,  
            31, 31, 30, 31, 30, 31  
        };  
        int total = 0;  
        // Days from previous years  
        for (int y = 1971; y < year; y++) {
```

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```

        if (isLeap(y))
            total += 366;
        else
            total += 365;
    }
    // Days from previous months
    for (int m = 1; m < month; m++) {
        total += days[m - 1];
    }
    if (m == 2 && isLeap(year)) {
        total++;
    }
    // Current month's days
    total += day;
    return total;
}

private boolean isLeap(int year) {
    return (year % 400 == 0) ||
        (year % 4 == 0 && year % 100 != 0);
}
}

```

Input: date1 = "2019-06-29", date2 = "2019-06-30"

Output: 1

Session 12

Date and Time Handling Concepts:

1. Day of the Year –

Java code-

```
class Solution {
    public int dayOfYear(String date) {
        int year = Integer.parseInt(date.substring(0, 4));
        int month = Integer.parseInt(date.substring(5, 7));
        int day = Integer.parseInt(date.substring(8, 10));
        int[] days = {
            31, 28, 31, 30, 31, 30,
            31, 31, 30, 31, 30, 31
        };
        int total = 0;
        for (int i = 0; i < month - 1; i++) {
            total += days[i];
        }
        // Leap year
        if (month > 2 && isLeapYear(year)) {
            total++;
        }
        total += day;
        return total;
    }
    private boolean isLeapYear(int year) {
```

```
return (year % 400 == 0) ||
```

```
        (year % 4 == 0 && year % 100 != 0);  
    }  
}
```

Input: date = "2019-01-09"

Output: 9

Explanation: Given date is the 9th day of the year in 2019.

2. Day of the Week –

Java code:

```
class Solution {  
    public String dayOfTheWeek(int day, int month, int year) {  
        String[] week = {  
            "Sunday",  
            "Monday",  
            "Tuesday",  
            "Wednesday",  
            "Thursday",  
            "Friday",  
            "Saturday"  
        };  
        int[] days = {  
            31, 28, 31, 30, 31, 30,  
            31, 31, 30, 31, 30, 31};  
        int total = 0;  
        // Count complete years from 1971  
        for (int y = 1971; y < year; y++) {  
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```

```

        if (isLeap(y)) {
            total += 366;
        } else {
            total += 365;
        }
    }

    // Count complete months
    for (int m = 1; m < month; m++) {
        total += days[m - 1];
        if (m == 2 && isLeap(year)) {
            total++;
        }
    }

    // Add current day
    total += day - 1;

    // January 1, 1971 = Friday
    int index = (5 + total) % 7
    return week[index];
}

private boolean isLeap(int year) {
    return year % 400 == 0 ||
        (year % 4 == 0 && year % 100 != 0);}

```

Input: day = 31, month = 8, year = 2019

Output: "Saturday"

Session 13-

Collection Processing & Transformation Operations-

1. Java Priority Queue –

Java code:

```
import java.util.*;

class Student {
    private int id;
    private String name;
    private double cgpa;
    public Student(int id, String name, double cgpa) {
        this.id = id;
        this.name = name;
        this.cgpa = cgpa;
    }
    public int getID() {
        return id;
    }
    public String getName() {
        return name;
    }
    public double getCGPA() {
        return cgpa; }
}

class Priorities {
    public List<Student> getStudents(List<String> events) {
```

```
PriorityQueue<Student> pq = new PriorityQueue<>()
```

```

        new Comparator<Student>() {
            public int compare(Student a, Student b) {
                if (a.getCGPA() != b.getCGPA()) {
                    return Double.compare(b.getCGPA(), a.getCGPA());
                }
            }
        }
        int nameCompare = a.getName().compareTo(b.getName());
        if (nameCompare != 0) {
            return nameCompare;
        }
        return Integer.compare(a.getID(), b.getID());
    }
}

);

for (String event : events) {
    String[] parts = event.split(" ");
    if (parts[0].equals("ENTER")) {
        pq.add(new Student(
            Integer.parseInt(parts[3]),
            parts[1],
            Double.parseDouble(parts[2])
        ));
    } else {
        if (!pq.isEmpty()) {
            pq.poll();
        }
    }
}
}

```

```
List<Student> result = new ArrayList<>();  
while (!pq.isEmpty()) {  
    result.add(pq.poll());  
}  
return result;  
}  
}
```

Sample Output -

Dan

Ashley

Shafaet

Maria

2. Java ArrayList –

Java code:

```
import java.io.*;
import java.util.*;

public class Solution{

    public static void main(String[] args){

        Scanner sc=new Scanner(System.in);

        int n=sc.nextInt();

        ArrayList<ArrayList<Integer>> list=new ArrayList<>();

        for(int i=0;i<n;i++){

            int d=sc.nextInt();

            ArrayList<Integer> row=new ArrayList<>();

            for(int j=0;j<d;j++) row.add(sc.nextInt());

            list.add(row);

        }int q=sc.nextInt();

        for(int i=0;i<q;i++){

            int x=sc.nextInt();

            int y=sc.nextInt();

            try{

                System.out.println(list.get(x-1).get(y-1));

            }catch(Exception e){

                System.out.println("ERROR!");

            }

        }sc.close();

    }

}
```

Output

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74

52

37

ERROR!

ERROR!

Session 14 –

Custom Comparators:

1. Largest Number –

Java code:

```
import java.util.*;

class Solution{

    public String largestNumber(int[] nums){

        String[] a=new String[nums.length];

        for(int i=0;i<nums.length;i++)a[i]=String.valueOf(nums[i]);

        Arrays.sort(a,(x,y)->(y+x).compareTo(x+y));

        if(a[0].equals("0"))return "0";

        StringBuilder s=new StringBuilder();

        for(String x:a)s.append(x);

        return s.toString();

    }

}
```

Input: nums = [10,2]

Output: "210"

Session 15-

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Custom Comparators & Sorting Logic:

1. Sort Characters by Frequency –

Java code:

```
import java.util.*;

class Solution{

    public String frequencySort(String s){
        Map<Character,Integer>m=new HashMap<>();
        for(char c:s.toCharArray())m.put(c,m.getOrDefault(c,0)+1);
        List<Character>a=new ArrayList<>(m.keySet());
        a.sort((x,y)->m.get(y)-m.get(x));
        StringBuilder r=new StringBuilder();
        for(char c:a)for(int i=0;i<m.get(c);i++)r.append(c);
        return r.toString();
    }
}
```

Input: s = "tree"

Output: "eert"

Explanation: 'e' appears twice while 'r' and 't' both appear once.

So 'e' must appear before both 'r' and 't'. Therefore "eetr" is also a valid answer.